

Part VI

Practice Papers

Practice Paper A — Quantitative Core

100 marks total. Full worked solutions follow both papers. Work each question by hand before checking the solution — the arithmetic is the point, not just the final number. A basic calculator is assumed; computing $\Phi(\cdot)$ or $\varphi(\cdot)$ to 4 decimal places from a standard table or approximation is expected.

Question 1 (10 marks). A SOFR discount curve gives $P(0, 4) = 0.8530$ and $P(0, 4.5) = 0.8365$. Using Actual/360 with $\tau = 0.5$ exactly, find the 6-month forward rate for the period $[4, 4.5]$. (5 marks for method, 5 for correct numerical answer.)

Question 2 (10 marks). Using discount factors $P(0, 1) = 0.9700$, $P(0, 2) = 0.9350$, $P(0, 3) = 0.8960$, $P(0, 4) = 0.8540$ with annual payments and $\tau_i = 1$ throughout, find the 4-year par swap rate and the annuity $A(0)$. (6 marks for the annuity, 4 for the par rate.)

Question 3 (12 marks). A caplet has forward $F = 3.95\%$, strike $K = 3.75\%$, fixing time $T_1 = 2.0$ years, accrual $\tau = 0.25$, discount factor to payment $P(0, T_2) = 0.9280$, notional \$25mm. Price it using Bachelier's formula with a normal volatility of $\sigma_n = 82$ bp/yr. Show d , $\Phi(d)$, $\varphi(d)$ and the final price.

Question 4 (8 marks). For the same caplet as Question 3, state the floorlet price *without* repeating the full Bachelier calculation, using cap–floor parity. Show your working.

Question 5 (12 marks). A 2y $\times$ 5y ATM payer swaption has annuity $A(0) = 4.35$, normal volatility $\sigma_n = 88$ bp/yr, expiry $T = 2$, notional \$30mm. Using the Bachelier ATM shortcut, find the premium. Show the rate-unit premium before converting to cash.

Question 6 (14 marks). Derive, from first principles using Itô's lemma applied to $\log(1 + \tau_j L_j(t))$, the volatility $\gamma_j(t)$ of the numeraire ratio $P(t, T_j)/P(t, T_{j+1})$, given that $dL_j = \sigma_j L_j dW_j^{j+1}$ under L_j 's own forward measure. State your answer in the form given in Chapter 13's key result box.

Question 7 (14 marks). A three-rate BGM system (terminal measure, $N = 3$) has, at some time t : $\tau = 0.5$ throughout, $L_0(t) = 3.60\%$, $L_1(t) = 3.80\%$, $L_2(t) = 3.95\%$, lognormal vols $\sigma_0 = 19\%$, $\sigma_1 = 21\%$, $\sigma_2 = 23\%$, and correlations $\rho_{01} = 0.88$, $\rho_{02} = 0.80$. Compute the full drift of $L_0(t)$ under the terminal measure (both contributing terms, $j = 1$ and $j = 2$).

Question 8 (10 marks). Using Rebonato's humped volatility form with $a = 0.04$, $b = 0.25$, $c = 1.00$, $d = 0.08$, evaluate $\sigma_i(t)$ for a rate with 2.5 years to expiry. State the long-run floor this form converges to as time-to-expiry $\rightarrow \infty$, without recomputing.

Question 9 (10 marks). A rank-reduction exercise: given that the first three eigenvalues of a 10×10 historical forward-rate correlation matrix are 7.35, 1.42, 0.61, and the remaining seven eigenvalues sum to 0.62, what fraction of total variance is explained by keeping only the first three factors? Would you recommend a 3-factor reduction on this evidence alone? Justify briefly.

